

# **STA 35C: Statistical Data Science III**

## **Lecture 6: Multiple Linear Regression & Polynomial Regression**

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## Recap: Simple linear regression

- Linear model:  $Y = \beta_0 + \beta_1 X + \varepsilon$
- *Least squares estimation* chooses  $(\hat{\beta}_0, \hat{\beta}_1)$  minimizing

$$\text{RSS}(b_0, b_1) = \sum_{i=1}^n (y_i - b_0 - b_1 x_i)^2$$

- *Interpretation:*  $\beta_1$  is the average change in the mean response per one-unit increase in  $X$ 
  - Associational, not necessarily causation
- *Inference:*  $\hat{\beta}_0$  and  $\hat{\beta}_1$  are random variables
  - Standard errors, confidence intervals, and tests
  - Rejecting  $H_0 : \beta_1 = 0$  gives evidence of a nonzero slope
- Model fit is summarized by RSE and

$$R^2 = 1 - \frac{\text{RSS}}{\text{TSS}} \quad \text{where} \quad \text{TSS} = \sum_{i=1}^n (y_i - \bar{y})^2$$

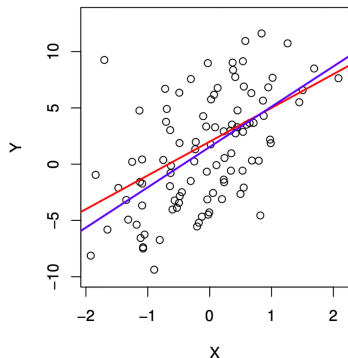


Figure: Least squares coefficient estimates from 100 data points.  
Red: population regression line,  
Blue: least squares line  
[JWHT21, Figure 3.3].

## Completing Lecture 5: The $R^2$ statistic

The mean-only prediction is  $\bar{y}$ ; the fitted (linear) regression model predicts  $\hat{y}_i$ .

$$\text{TSS} = \sum_{i=1}^n (y_i - \bar{y})^2, \quad \text{RSS} = \sum_{i=1}^n (y_i - \hat{y}_i)^2, \quad R^2 = 1 - \frac{\text{RSS}}{\text{TSS}}.$$

- $R^2$  is the proportion of observed variation captured by the fitted model
- For least squares with an intercept and  $\text{TSS} > 0$ , training  $R^2 \in [0, 1]$  and is scale-free
- High training  $R^2$  does not guarantee a significant slope, accurate individual predictions, or good test performance

### Example (Four observations)

Data  $(x_i, y_i)$ :  $(1, 1), (1, 3), (2, 3), (2, 5)$ , so  $\bar{y} = 3$

Least squares:  $\hat{y} = 2x$ , giving fitted values 2, 2, 4, 4

$\text{TSS} = 4 + 0 + 0 + 4 = 8$ ,  $\text{RSS} = 1 + 1 + 1 + 1 = 4$

$R^2 = 1 - \frac{4}{8} = \frac{1}{2}$ : the line captures 50% of the observed variation

# Agenda

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## Last time: Simple linear regression

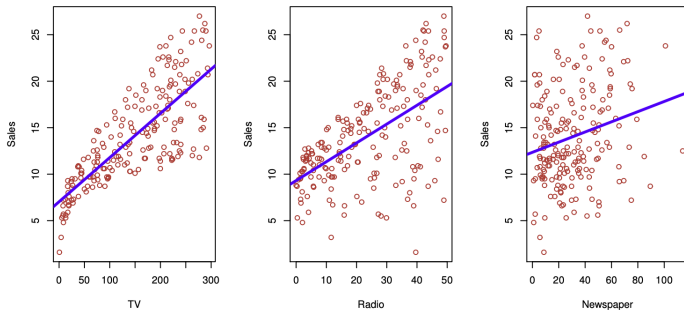
- Model, least squares, inference, and model fit

## Today: Extending simple linear regression

- Multiple linear regression
  - Motivation: *"What if we have more than one predictor?"*  $X_1, X_2, \dots$
  - Model
  - Estimation
  - Inference
  - Model fit
  - Prediction intervals
- Polynomial regression
  - Motivation: *"What if  $X_1$  and  $X_2$  interact, or if  $Y$  depends on  $X^2$  instead of  $X$ ?"*

# Motivation for multiple linear regression

Recall the **Advertising** dataset and the three separate simple linear regression lines:



**Figure:** The **Advertising** data set: **Sales** of a product in 200 different markets against advertising budgets for three media: **TV**, **Radio**, and **Newspaper** [JWHT21, Figure 2.1].

**Problem:** Each simple linear regression line ignores the other two predictors

**Question:** Can we extend our analysis to accommodate *all* predictors simultaneously?

# Multiple linear regression: Setup

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We predict  $Y$  using multiple variables  $X_1, X_2, \dots, X_p$ , assuming:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_p X_p + \varepsilon,$$

- *Model parameters:*  $\beta_0, \beta_1, \dots, \beta_p$  are fixed, unknown constants
- $\varepsilon$  is an error term, satisfying  $\mathbb{E}[\varepsilon \mid X_1, \dots, X_p] = 0$

The coefficient  $\beta_j$  is interpreted as the **average change** in the **mean response** for a one-unit increase in  $X_j$ , **holding all other predictors fixed**

- This is an associational interpretation, not necessarily a causal one

Once we *estimate*  $\hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_p$  from training data, we can predict

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \dots + \hat{\beta}_p x_p$$

# Visualizing multiple linear regression

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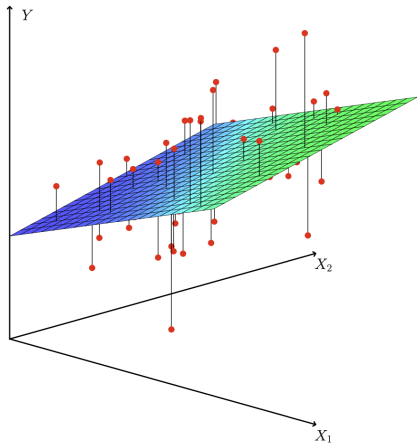


Figure: An illustration of multiple linear regression [JWHT21, Figure 3.4]. With two predictors, the fitted surface is a plane; with many predictors, it is a hyperplane.

## Coefficient estimation via least squares

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Coefficients  $\beta_0, \beta_1, \dots, \beta_p$  must be estimated from data  $(x_1, y_1), \dots, (x_n, y_n)$ , where

$$x_i = (x_{i1}, x_{i2}, \dots, x_{ip}).$$

Again, we use the **least squares** criterion:

- The *least squares* approach chooses  $\hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_p$  that minimize the RSS:

$$\text{RSS}(b_0, \dots, b_p) = \sum_{i=1}^n (y_i - b_0 - b_1 x_{i1} - \dots - b_p x_{ip})^2$$

- (Optional) In matrix form<sup>1</sup>, when  $\mathbf{X}^\top \mathbf{X}$  is invertible:

$$\hat{\beta} = \begin{bmatrix} \hat{\beta}_0 \\ \vdots \\ \hat{\beta}_p \end{bmatrix} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y} \quad \text{where} \quad \mathbf{X} = \begin{bmatrix} 1 & x_{11} & \cdots & x_{1p} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & \cdots & x_{np} \end{bmatrix} \quad \text{and} \quad \mathbf{y} = \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix}$$

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<sup>1</sup>This can be derived by setting the partial derivatives of RSS to 0, but is beyond the scope of STA 35C

## Pop-up quiz: Interpreting coefficients

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**Scenario:** We fit a multiple linear regression model on the **Advertising** dataset:

$$\text{Sales} = \beta_0 + \beta_1 \text{TV} + \beta_2 \text{Radio} + \beta_3 \text{Newspaper} + \varepsilon.$$

Suppose that we obtain:  $\hat{\beta}_1 = 0.04$ ,  $\hat{\beta}_2 = 0.18$ ,  $\hat{\beta}_3 = -0.02$ .

**Question:** Which statement *best* describes the meaning of  $\hat{\beta}_1 = 0.04$  in this model?

**Multiple-choice answers:**

- A) TV advertising alone explains 4% of the variation in **Sales**
- B) A 1-unit increase in **TV** causes **Sales** to increase by exactly 0.04
- C) Holding **Radio** and **Newspaper** fixed, a 1-unit increase in **TV** is associated with an average increase of 0.04 units in predicted **Sales**
- D) The effect of **TV** is 0.04 only when **Radio** and **Newspaper** are both 0

**Answer: C.**

In multiple regression, a coefficient measures the average change in the predicted response for a one-unit increase in that predictor, holding the others fixed

## Some key questions with multiple predictors

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When we perform multiple linear regression, we often want to answer questions like:

- Are predictors  $X_1, \dots, X_p$  related to  $Y$  (i.e., do they help predict  $Y$ )?
- Which subset of  $X_1, \dots, X_p$  is most important?
- How well does the model fit the data?
- Given new predictor values, what response value should we predict and how accurate is that prediction?

Let's address these questions one by one

## Hypothesis testing for relationship between $Y$ and each $X_j$

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Recall from simple linear regression:

$$H_0 : \beta_1 = 0 \quad \text{vs.} \quad H_1 : \beta_1 \neq 0, \quad t = \frac{\hat{\beta}_1}{\widehat{\text{SE}}(\hat{\beta}_1)} \sim t_{n-2} \text{ under } H_0.$$

- **Idea:** If  $H_0$  is true,  $t$  follows the t-distribution and will likely be close to 0
- Thus, a very large  $|t|$  value is an evidence against  $H_0 \rightarrow$  reject  $H_0$

In multiple linear regression, we do the same, but for each predictor  $X_j$ :

$$H_0 : \beta_j = 0 \quad \text{vs.} \quad H_1 : \beta_j \neq 0, \quad t = \frac{\hat{\beta}_j}{\widehat{\text{SE}}(\hat{\beta}_j)} \sim t_{n-p-1} \text{ under } H_0.$$

- Large  $|t_j|$  is evidence against  $H_0$ ; use the critical value for the chosen significance level
- This tests a zero partial slope ( $Y$  vs  $X_j$ ) *after accounting for the other predictors*
- The answer can differ from simple regression setting when predictors are correlated
- Also, we may instead ask whether *any* predictor is useful at all (will see shortly)

## Advertising example: Simple vs. multiple regressions

Q: Is **newspaper** associated with **sales**, before and after adjusting for other media?

|           | Coefficient | Std. error | t-statistic | p-value  |
|-----------|-------------|------------|-------------|----------|
| Intercept | 7.0325      | 0.4578     | 15.36       | < 0.0001 |
| TV        | 0.0475      | 0.0027     | 17.67       | < 0.0001 |

|           | Coefficient | Std. error | t-statistic | p-value  |
|-----------|-------------|------------|-------------|----------|
| Intercept | 9.312       | 0.563      | 16.54       | < 0.0001 |
| radio     | 0.203       | 0.020      | 9.92        | < 0.0001 |

|           | Coefficient | Std. error | t-statistic | p-value  |
|-----------|-------------|------------|-------------|----------|
| Intercept | 12.351      | 0.621      | 19.88       | < 0.0001 |
| newspaper | 0.055       | 0.017      | 3.30        | 0.00115  |

Figure: Separate simple regressions suggest **TV**, **radio**, and **newspaper** are all significant [JWHT21, Tables 3.1 & 3.3].

|           | Coefficient | Std. error | t-statistic | p-value  |
|-----------|-------------|------------|-------------|----------|
| Intercept | 2.939       | 0.3119     | 9.42        | < 0.0001 |
| TV        | 0.046       | 0.0014     | 32.81       | < 0.0001 |
| radio     | 0.189       | 0.0086     | 21.89       | < 0.0001 |
| newspaper | -0.001      | 0.0059     | -0.18       | 0.8599   |

Figure: Multiple regression suggests **newspaper** is not significant [JWHT21, Table 3.4].

|           | TV     | radio  | newspaper | sales  |
|-----------|--------|--------|-----------|--------|
| TV        | 1.0000 | 0.0548 | 0.0567    | 0.7822 |
| radio     |        | 1.0000 | 0.3541    | 0.5762 |
| newspaper |        |        | 1.0000    | 0.2283 |
| sales     |        |        |           | 1.0000 |

Figure: Correlation matrix for **TV**, **radio**, **newspaper**, and **sales** [JWHT21, Table 3.5].

**Takeaway:** In multiple regression,  $\beta_j$  measures the association between  $X_j$  and  $Y$  *holding the other predictors fixed*; correlated predictors can therefore change significance

## Advertising example: Testing single vs. any predictor

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Q: Is “any” (=at least one) of TV, radio, newspaper useful in predicting sales?

We test the *joint* hypothesis:

$$H_0 : \beta_1 = \beta_2 = \cdots = \beta_p = 0 \quad \text{vs.} \quad H_1 : \text{at least one } \beta_j \neq 0$$

At level  $\alpha$ , use the *F-statistic* and its  $(1 - \alpha)$  quantile:

$$F = \frac{(\text{TSS} - \text{RSS})/p}{\text{RSS}/(n - p - 1)} \quad \begin{cases} \text{Reject } H_0 & \text{if } F > F_{p, n-p-1; 1-\alpha}, \\ \text{Do not reject } H_0 & \text{otherwise.} \end{cases}$$

**Rationale:** If  $H_0$  is true, both mean squares in the ratio estimate  $\sigma^2$

- If all predictors are useless, adding them should not reduce RSS much, so  $F$  should be near 1
- A large  $F$  means the RSS reduction is larger than expected from noise alone
- Under  $H_0$ ,  $F$  follows an  $F$ -distribution with  $(p, n - p - 1)$  degrees of freedom

We will discuss how to select “important” predictors later in this course (after Midterm 1)

## Evaluating the model fit

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Training fit of a multiple linear regression<sup>2</sup> can be measured by the RSE or the  $R^2$

- **Residual standard error (RSE):** “*typical size of deviation, in the same units as  $Y$* ”

$$\text{RSE} = \sqrt{\frac{\text{RSS}}{n - p - 1}} \quad \text{where} \quad \text{RSS} = \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

- **The  $R^2$ :** “the proportion of observed variation captured by the fitted model”

$$R^2 = \frac{\text{TSS} - \text{RSS}}{\text{TSS}} = 1 - \frac{\text{RSS}}{\text{TSS}}, \quad \text{where} \quad \text{TSS} = \sum_{i=1}^n (y_i - \bar{y})^2$$

- **Note:** On the *same observations*, adding predictors cannot reduce training  $R^2$
- **“Adjusted”  $R^2$**  compensates for adding predictors:

$$R_{\text{adj}}^2 = 1 - \frac{\text{RSS}/(n - p - 1)}{\text{TSS}/(n - 1)}$$

- It rises only when the added predictors improve fit to justify the extra complexity

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<sup>2</sup>Full-rank model,  $n > p + 1$ ,  $\text{TSS} > 0$

## Pop-up quiz: $R^2$ vs. adjusted $R^2$

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**Scenario:** With  $n = 22$  observations, suppose a one-predictor model with an intercept has

$$R^2 = 0.80, \quad R^2_{\text{adj}} = 0.79.$$

After adding a second predictor  $X_2$  (suspected to be mostly noise), we get:

$$R^2 = 0.805, \quad R^2_{\text{adj}} \approx 0.784.$$

**Question:** Why did  $R^2$  go up while  $R^2_{\text{adj}}$  went down?

- A) There must be a calculation error; if  $R^2$  increases,  $R^2_{\text{adj}}$  must also increase.
- B)  $X_2$  reduced RSS, but not enough to offset losing one residual degree of freedom; adjusted  $R^2$  decreased
- C) Adjusted  $R^2$  always decreases whenever we add a predictor, no matter how useful it is
- D)  $R^2$  does not measure model fit at all, whereas  $R^2_{\text{adj}}$  is the only valid measure of fit

**Answer: B.**

Here,  $1 - (1 - 0.805) \frac{21}{19} \approx 0.784 < 0.79$ . This is a training-fit comparison, not proof that  $X_2$  contains no signal

# Confidence intervals and prediction intervals

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With  $\hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_p$ , we predict  $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \dots + \hat{\beta}_p x_p$

Under the stated model assumptions, consider two kinds of uncertainty:

- $\hat{y} = \hat{f}_{\hat{\beta}}(x)$  only *estimates*  $f_{\beta}(x) = \beta_0 + \sum_{j=1}^p \beta_j x_j$  → variability due to estimates  $\hat{\beta}_j$
- $y = f(x) + \varepsilon$  has an error term → additional variability due to noise

## Confidence interval for the mean response $f(x)$ :

- Reflects uncertainty in prediction due to estimating coefficients
- Across repeated samples, 95% of CIs would contain the true *mean response* at  $x$

## Prediction interval for a new response $Y_{\text{new}} \mid X = x$ :

- Accounts for uncertainty in both  $\hat{y} = \hat{f}(x)$  *and* the random noise  $\varepsilon$
- Over repeated training samples and new observations at fixed  $x$ , 95% of PIs cover  $Y_{\text{new}}$
- For the same fit,  $x$ , and confidence level, a PI is wider than a mean-response CI

In **R** (`x0`: a data frame with matching predictor names):

```
predict(model, newdata = x0, interval = "confidence", level = 0.95)
```

```
predict(model, newdata = x0, interval = "prediction", level = 0.95)
```

# What if there is a non-linear relationship?

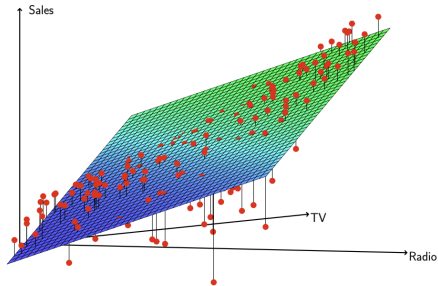


Figure: Pronounced synergy between **TV** and **Radio**; positive residuals cluster along the 45-degree line [JWHT21, Figure 3.5].

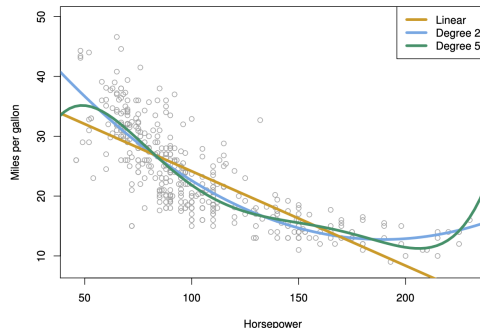


Figure: A non-linear relationship between **mpg** and **horsepower** is noticeable [JWHT21, Figure 3.8].

→ We can enrich the model by adding **interaction** terms (**TV**  $\times$  **Radio**) or **non-linear** terms (**horsepower**<sup>2</sup>) to capture these effects

# Polynomial regression: Higher-order and interaction terms

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**Polynomial regression** extends the linear model by including powers of predictors<sup>3</sup>:

$$Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \dots + \beta_d X^d + \varepsilon$$

- Treated as multiple linear regression on transformed predictors  $(X, X^2, \dots, X^d)$
- Although non-linear in  $X$ , the model is still linear in the coefficients  $\beta_j$

**Example:** Interaction effect (synergy between **TV** and **Radio**)

$$\begin{aligned}\text{Sales} &= \beta_0 + \beta_1 \text{TV} + \beta_2 \text{Radio} + \beta_3 \underbrace{\text{TV} \times \text{Radio}}_{\text{interaction term}} + \varepsilon \\ &= \beta_0 + (\beta_1 + \beta_3 \text{Radio}) \text{TV} + \beta_2 \text{Radio} + \varepsilon\end{aligned}$$

- Holding **Radio** fixed, the TV slope for mean sales is  $\beta_1 + \beta_3 \text{Radio}$ , not  $\beta_1$  alone

**Example:** Quadratic model

$$\text{mpg} = \beta_0 + \beta_1 \text{horsepower} + \beta_2 \underbrace{\text{horsepower}^2}_{\text{nonlinear term}} + \varepsilon$$

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<sup>3</sup>More generally,  $Y = \sum_{\alpha: |\alpha| \leq d} \beta_{\alpha} \mathbf{X}^{\alpha} + \varepsilon$  where  $\alpha = (\alpha_1, \dots, \alpha_p)$  and  $\mathbf{X}^{\alpha} = X_1^{\alpha_1} X_2^{\alpha_2} \dots X_p^{\alpha_p}$

# Model flexibility and the risk of overfitting

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A more flexible model can fit the training data more closely

- For least squares on the same observations, adding terms cannot increase training RSS
- Training RSS = RSS measured on the training data

But a better fit on the training data does not necessarily mean a better model

- Our actual goal is good performance on new, unseen data
- A model that fits the training set extremely well may perform worse on a test set

**Overfitting:** If a model is too flexible, it can fit noise as well as signal

- Typical symptom: very small training error, but noticeably larger test error
- Compare training fit with performance on new data, not with training fit alone
- We will return to this issue repeatedly throughout the course

## Wrap-up

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Multiple linear regression extends simple linear regression to several predictors:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_p + \varepsilon$$

- Each slope coefficient  $\beta_j$  describes the change in the mean response per unit increase in  $X_j$ , holding the other predictors fixed
- Individual  $t$ -tests and the overall  $F$ -test answer different questions: Association to one predictor vs. any predictor
- Model fit is summarized by RSE,  $R^2$ , and adjusted  $R^2$ ; adjusted  $R^2$  helps compare models with different numbers of predictors
- Mean-response confidence intervals target  $f(x)$ ; prediction intervals target a new observation
- Interaction and polynomial terms make the model more flexible, but extra flexibility increases the risk of overfitting

**Suggested reading:** [JWHT21, Ch. 3.2, 3.3.2; optional Sec. 7.1]

**Next lecture:** Qualitative predictors and pitfalls in linear regression

# References

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Gareth James, Daniela Witten, Trevor Hastie, and Robert Tibshirani.

*An Introduction to Statistical Learning: with Applications in R*, volume 112 of *Springer Texts in Statistics*.

Springer, New York, NY, 2nd edition, 2021.